# LAB IN A CUBE!

# A PocketQube for Education

User's Manual

instituto de telecomunicações

# WHAT IS A POCKETQUBE ?

## SIZE

A PocketQube is a miniaturized satellite measuring exactly 5x5x5 cm

## HOW MANY ARE THERE?

There are currently about 37 standard PocketQubes in orbit as of mid-2026.

## SPACE FOR EVERYONE

PocketQubes were introduced in 2009 to lower the cost of accessing space. By shrinking satellite size and launch expenses, they opened the door for students, researchers, startups, and amateur engineers to reach orbit.

# WHAT IS

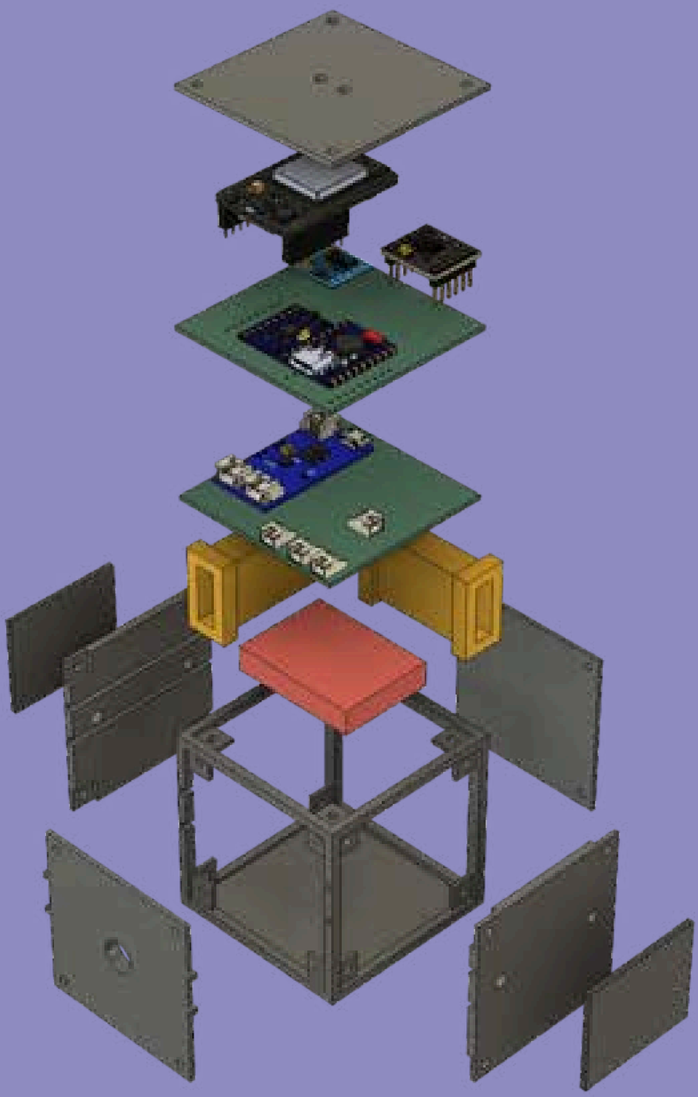
# OUR MISSION?

## EVERY SATELLITE HAS A MISSION

Our mission is to bridge the gap between space exploration interest and practical education by making real satellite technology accessible to everyone.

Through Lab in a Cube, a low-cost, "Do it yourself" PocketQube satellite kit, we empower students from secondary school to graduate level with hands-on laboratory experience in satellite subsystems, modular architecture, and attitude control.

By engineering a high-fidelity educational tool that costs approximately 85€ instead of the thousands charged for commercial kits, we are breaking down financial barriers to deliver an affordable, beginner-friendly, yet challenging gateway to the next generation of space innovators.



# HOW DOES IT WORK?

## COMMAND AND DATA HANDLING (CDH)

Serves as the satellite's central computing and data management unit. It processes commands received from the ground, coordinates subsystem operations, collects and stores mission data, and manages onboard software tasks.

## COMMUNICATIONS

Provides the link between the satellite and ground stations. It handles the transmission of telemetry, scientific data, and payload information while receiving commands from mission operators. Reliable communication ensures continuous monitoring, control, and data delivery throughout the satellite's mission.

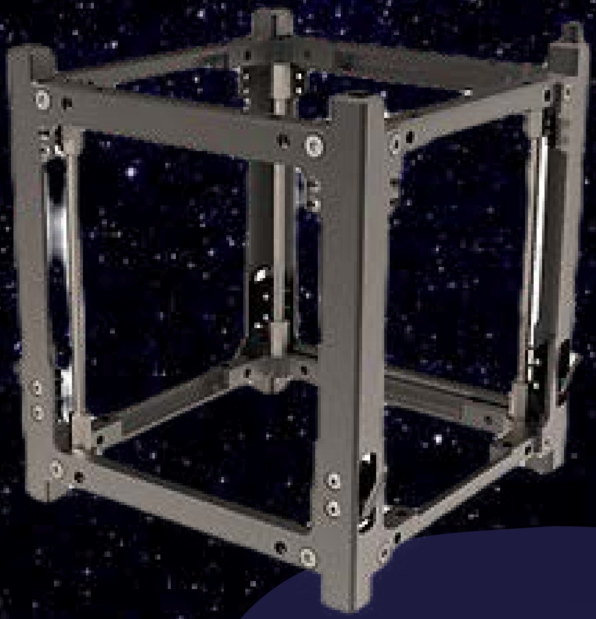
# HOW DOES IT WORK?

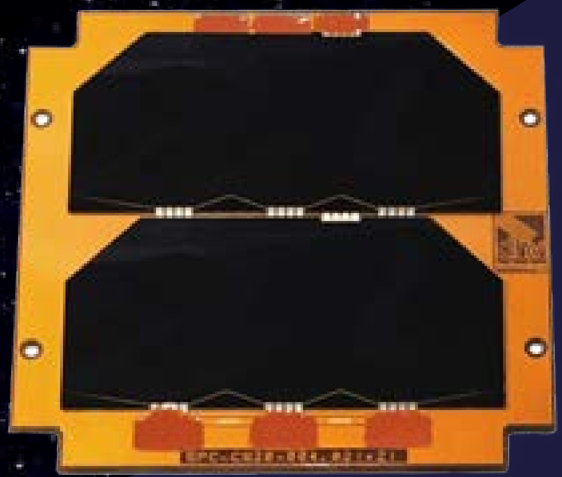
## STRUCTURES

Provides the mechanical framework that supports and protects all satellite components during launch and operation in space. It is designed to withstand launch loads, vibrations, and the harsh space environment while maintaining the alignment and integrity of onboard systems throughout the mission lifetime.

## ELECTRICAL POWER SYSTEM (EPS)

Generates, stores, and distributes electrical power throughout the satellite. Solar panels convert sunlight into electricity, while batteries provide energy during eclipse periods when sunlight is unavailable.





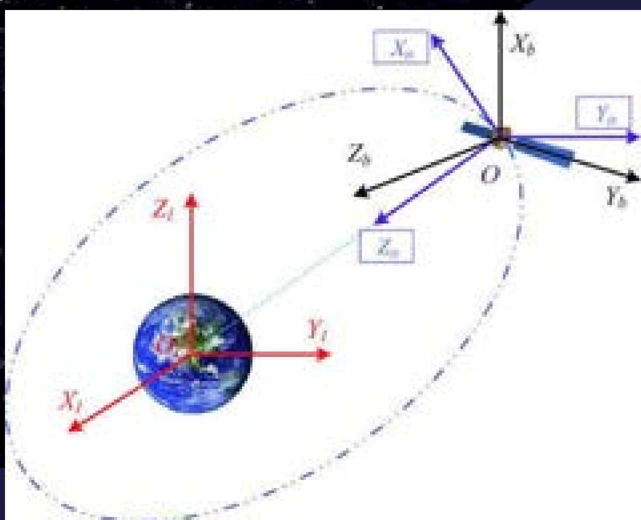
# HOW DOES IT WORK?

## PAYLOAD

This subsystem is the primary mission equipment carried by the satellite and is responsible for accomplishing the mission objectives. Depending on the mission, the payload may include cameras, scientific instruments, communication transponders, or sensors that collect, process, and transmit valuable data

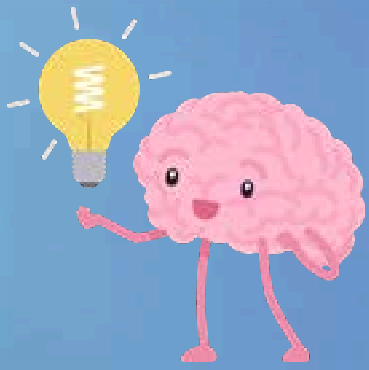
## ATTITUDE DETERMINATION AND CONTROL SYSTEM (ADCS)

Ensures the satellite maintains the correct orientation in space. Using sensors, magnetorquers, and control algorithms, it determines the spacecraft's attitude and performs precise maneuvers to point antennas, solar panels, and payloads toward their intended targets, enabling stable and efficient mission operations.




Wait!
What are this?
What am I supposed
to do with all this?

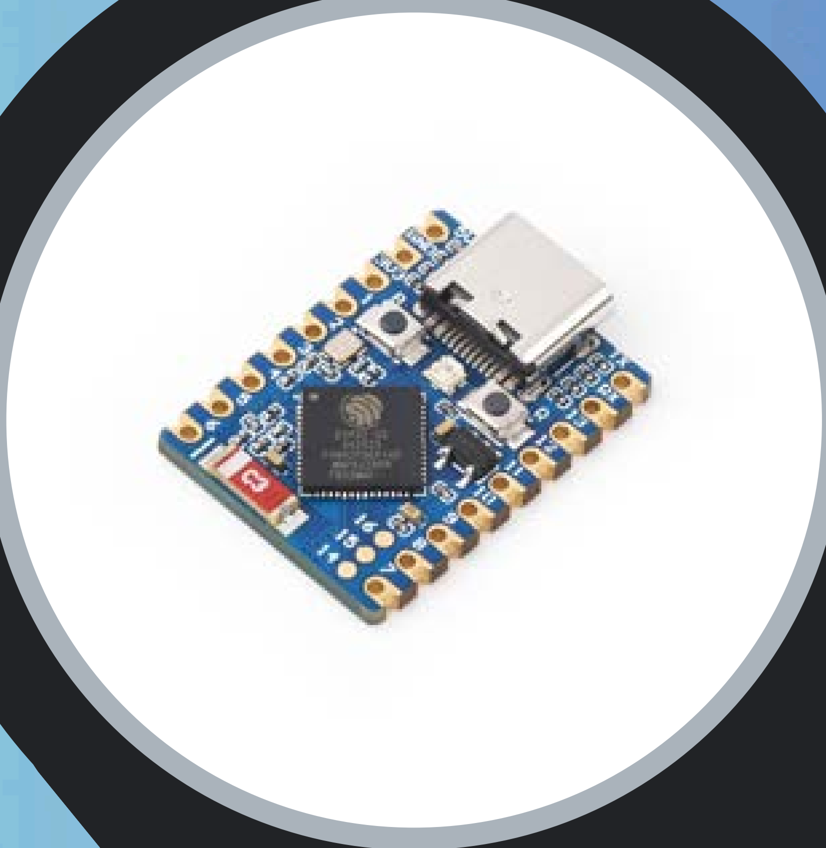
# ESP32 S3 Mini



A.K.A: The **Brains** of our PocketQube

It might look small, but this is the **ESP32 S3 Mini**—and it is the actual "super-brain" of our mission!

- **Invisible Connections:** It has built-in **Wi-Fi** and **Bluetooth**. This means it can talk to your phone, computer, or other devices completely wirelessly.
- **Pocket-Sized Power:** It is around the size of a 1 euro coin and **ultra-lightweight**, which is exactly what we need to fit into our PocketQube.
- **Sensor Friendly:** Those gold holes on the sides act like electronic plugs. You can easily connect "eyes" (like cameras) or "feelers" (like temperature sensors) to it.



# ESP32 CAM

A.K.A: The **Eyes** of our PocketQube

Every space explorer needs to see where it's going! This is the ESP32-CAM, and it acts as the official "eyes" of our PocketQube.

- **A Tiny Camera:** It comes with a miniature camera lens attached right to the board. It can observe the Earth just like a real satellite in space.

- **Photo Storage:** On the back, it has a slot for a tiny MicroSD card. This lets it save thousands of photos directly onto the board.

- **“Nap Time”:** Space satellites have limited battery power. This board has a "Deep Sleep" mode, meaning it can completely turn itself off to save energy and wake up only when it's time to snap a photo.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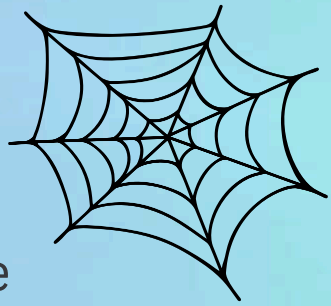
# Magnetometer



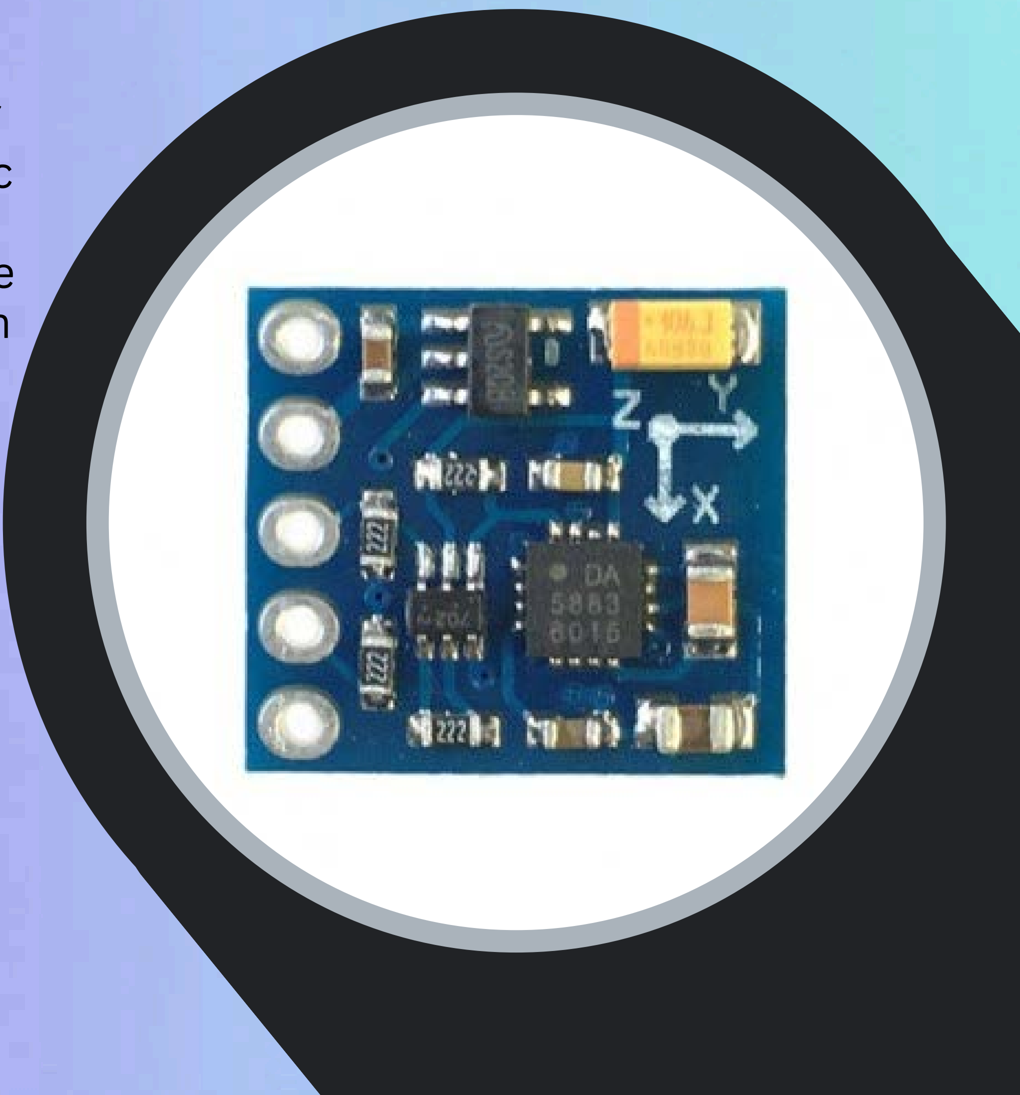
A.K.A:  the **Spider Sense** of our PocketQube

Just like Spider-Man feels a tingle when something invisible is happening around him, this sensor can feel invisible magnetic forces in space.

- **Sensing the Unseen:** Humans can't see or feel magnetic fields, but this sensor can! It feels the Earth's giant, invisible magnetic shield wrapping around the planet.

- **Metal & Danger Detector:** If a satellite gets too close to strong magnetic fields or metallic objects that could mess up its instruments, this sensor sniffs them out instantly.

- **Finding True North:** It acts like a compass, tracking the invisible lines of the Earth's magnetic web so it always knows exactly where the North and South poles are.

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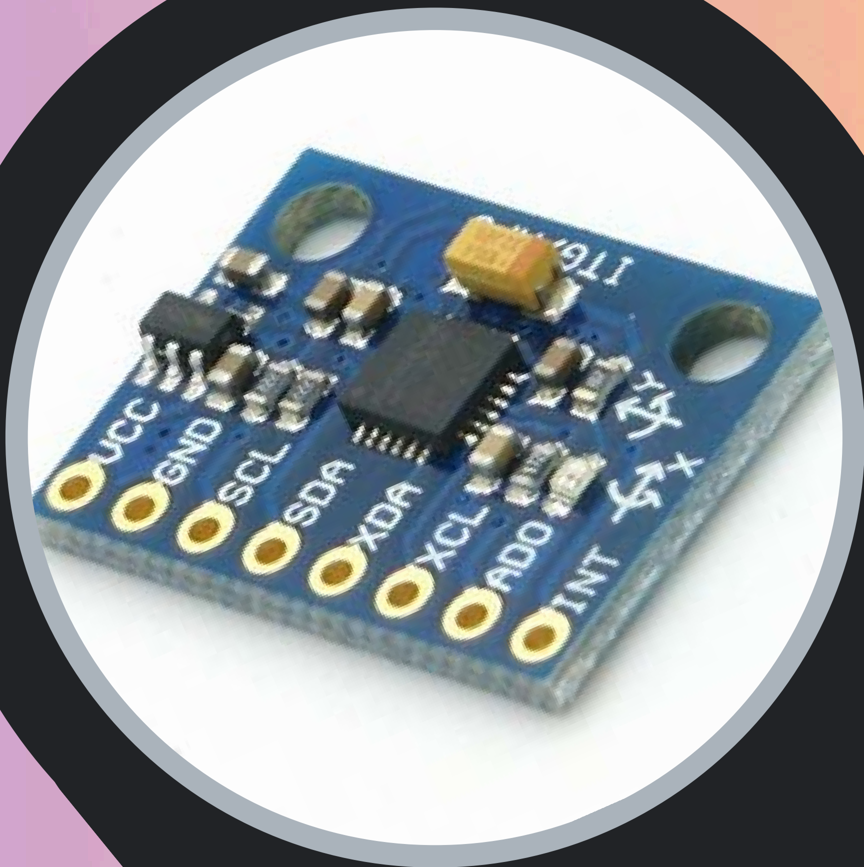
# IMU

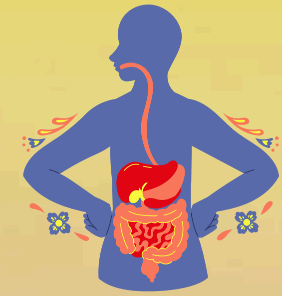
A.K.A: The **Sense of Direction** of our PocketQube

In the dark, floating in the void of space, there is no "up" or "down." Without this sensor, our satellite would get completely dizzy and lost.

- **Gyroscope:** It detects spinning: If a satellite starts tumbling around, the IMU feels it instantly so a satellite can track its spins.
- **Accelerometer:** It senses changes in speed and movement, when it enters zero-gravity, or if it tilts to the side.
- **Orientation:** The IMU tells a satellite's exact angle, whether it is facing Earth, looking out into deep space, or flying sideways.



# Battery Charger



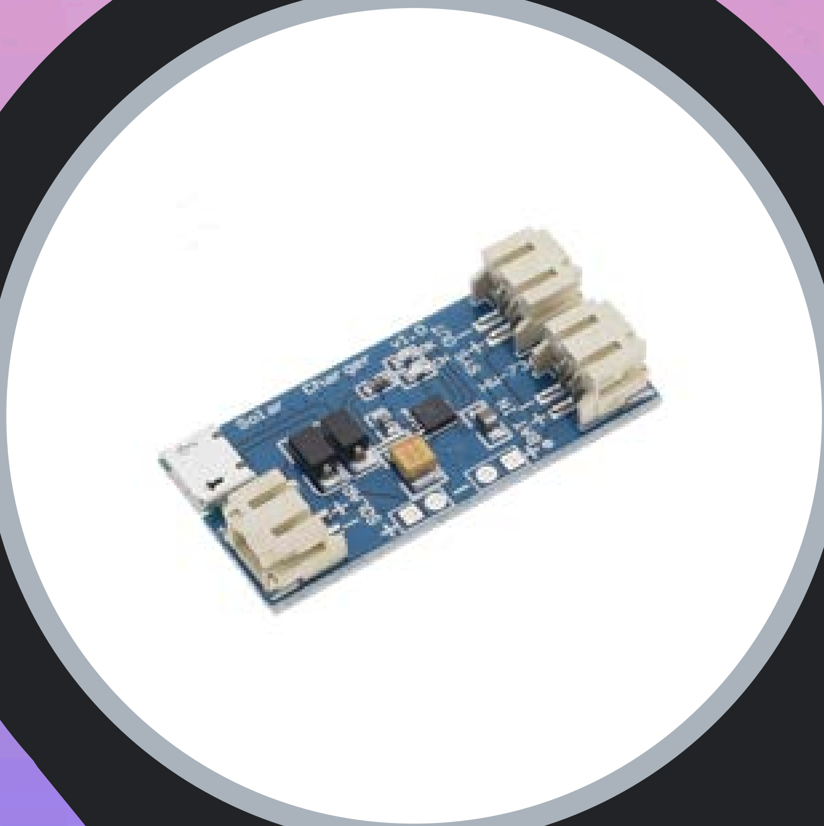
A.K.A:  The **Digestive System** of our PocketQube

Just as your digestive system processes food and sends nutrients to where they are needed, the battery charger processes the energy coming from the solar panels and delivers it safely to the battery.

- **Directing the Energy Flow**: The charger receives electricity from the solar panels and decides how much of it should be sent to the battery.

- **Preparing for Darkness:** Before a satellite enters Earth's shadow, the charger helps store energy in the battery so there is power available when the solar panels can't generate electricity.

- **Protecting the Battery:** Batteries need to be charged carefully. Too much voltage or current can damage them. The charger makes sure the battery receives power safely.



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# H-Bridge

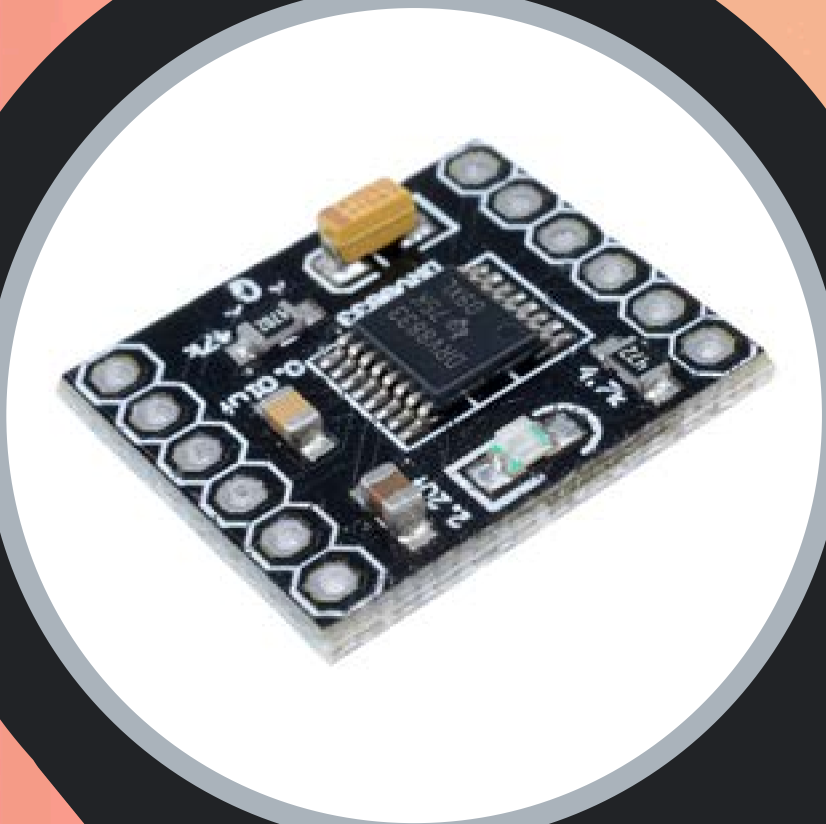
A.K.A:  The **Muscles** of our PocketQube

The H-bridge switches the direction of the current in the coils to change which way a satellite moves.

- **Direction Flipper:** The H-bridge acts like a “high-speed train track switch”, instantly flipping the current forward or backward to change a satellite's spin from clockwise to counter-clockwise.

- **Power Booster:** The H-bridge handles the heavy electrical currents, acting like a strength booster.

- **Brakes:** It doesn't just make a satellite move, it can also safely block the circuit to act as an electronic brake, stopping a satellite from spinning out of control.



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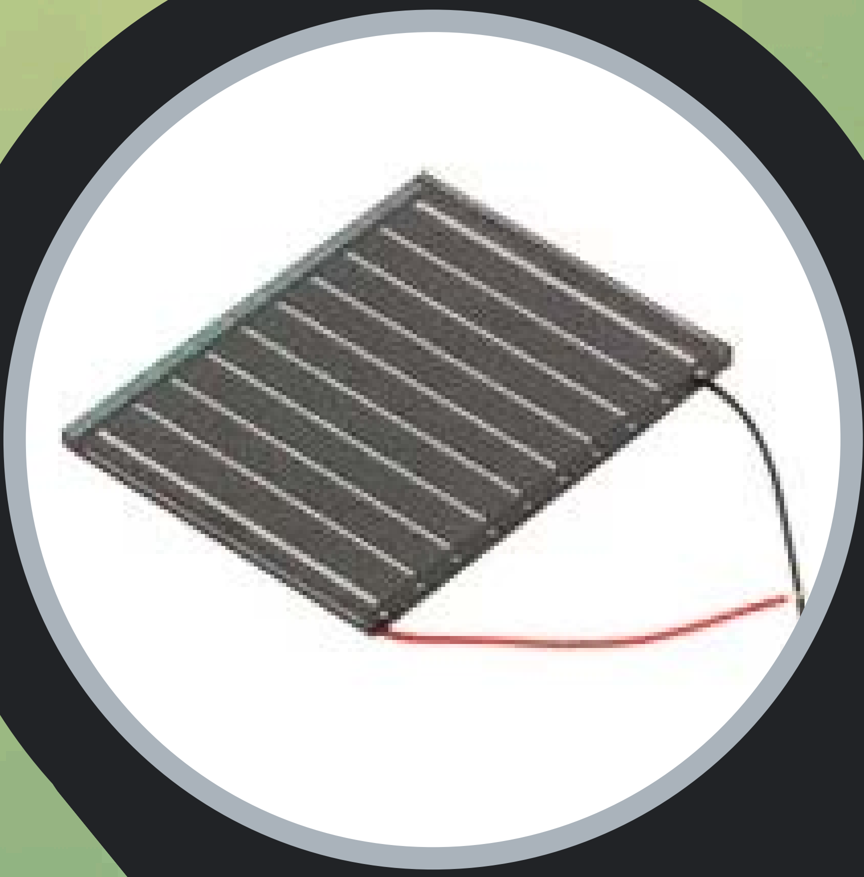
# Solar Panels

A.K.A: The **Skin** of our PocketQube

Just like our skin receives sunlight and warmth from the Sun, solar panels receive sunlight and turn it into energy to power a real satellite.

- **Essential for Survival:** Without solar panels, the PocketQube would run out of energy, just like a person would run out of energy without food.
- **Charging the Battery:** When the panels generate more power than the satellite needs, the extra energy is stored in a battery for use when the satellite passes through Earth's shadow.
- **Built for Efficiency:** Every square centimetre of a PocketQube is valuable, so the solar panels are carefully designed to capture as much energy as possible from the available sunlight.



# LiPo Battery

A.K.A:  The **Heart** of our PocketQube

Just like your heart beats to pump life through your body, this battery constantly pumps electrical power through the wires to keep every single component alive.

- **A Rechargeable Heart:** It doesn't die when it runs out of juice. It can be recharged hundreds of times, either by plugging it into a computer on Earth or using solar panels out in space!
- **Built-in Safety Bodyguards:** It works closely with protection chips that act like a immune system, stopping the battery from getting too hot, overcharging, or running completely empty.
- **Steady & Smooth Flow:** It delivers a perfectly smooth stream of electricity. This ensures the satellite's "brain" gets a steady heartbeat of power and doesn't suddenly freeze or glitch out.

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# Magnetorquers

A.K.A: The **Joints** of our PocketQube

They may look like simple coils of wire, but magnetorquers help the satellite change its orientation in space. By creating magnetic fields, they can gently rotate the PocketQube and help it point in the right direction.

- **Creating a Magnetic Field:** When electricity flows through the coils, they become tiny electromagnets.
- **Working with Earth's Magnetic Field:** The magnetic field produced by the coils interacts with Earth's magnetic field, creating small forces that can rotate the satellite.
- **No Fuel Required:** Unlike rockets or thrusters, they only need electrical power to operate.

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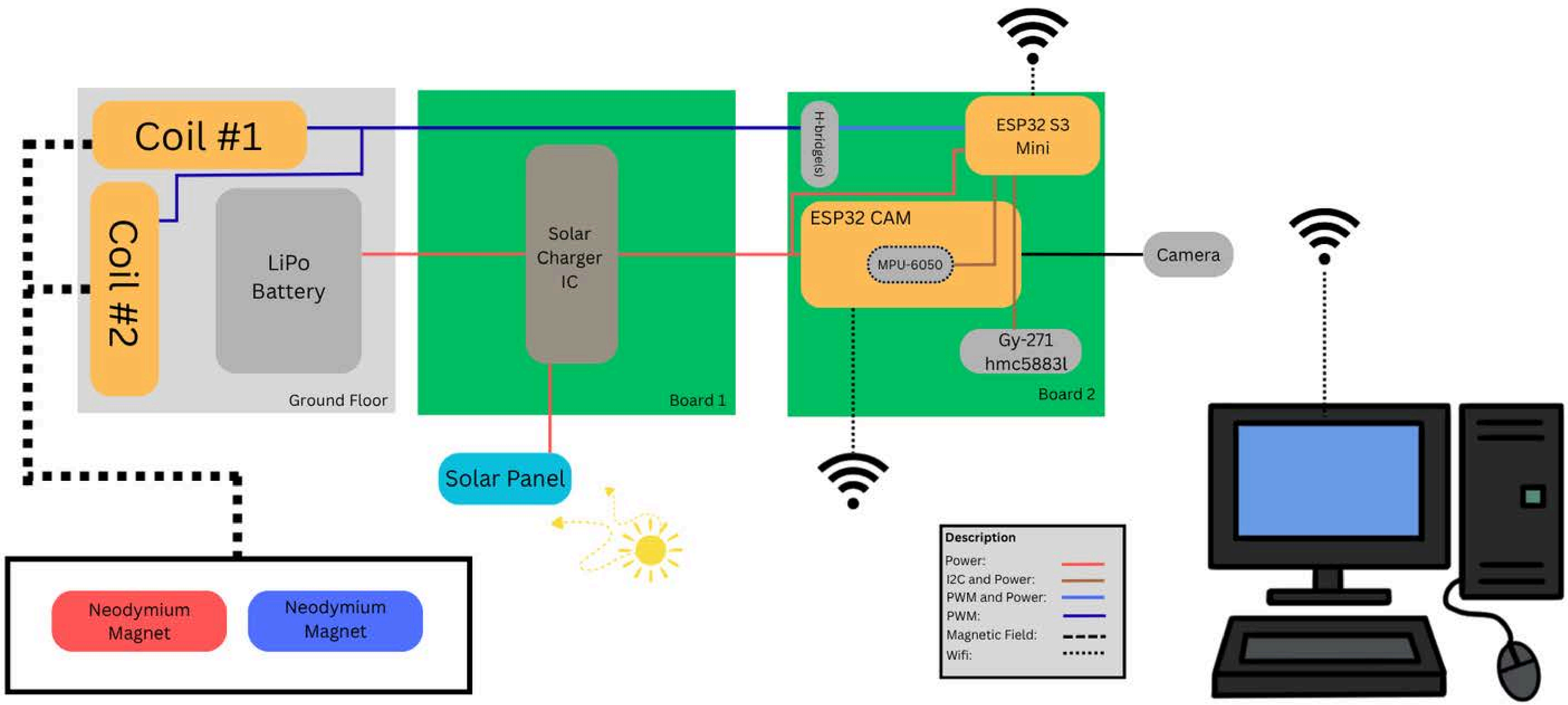
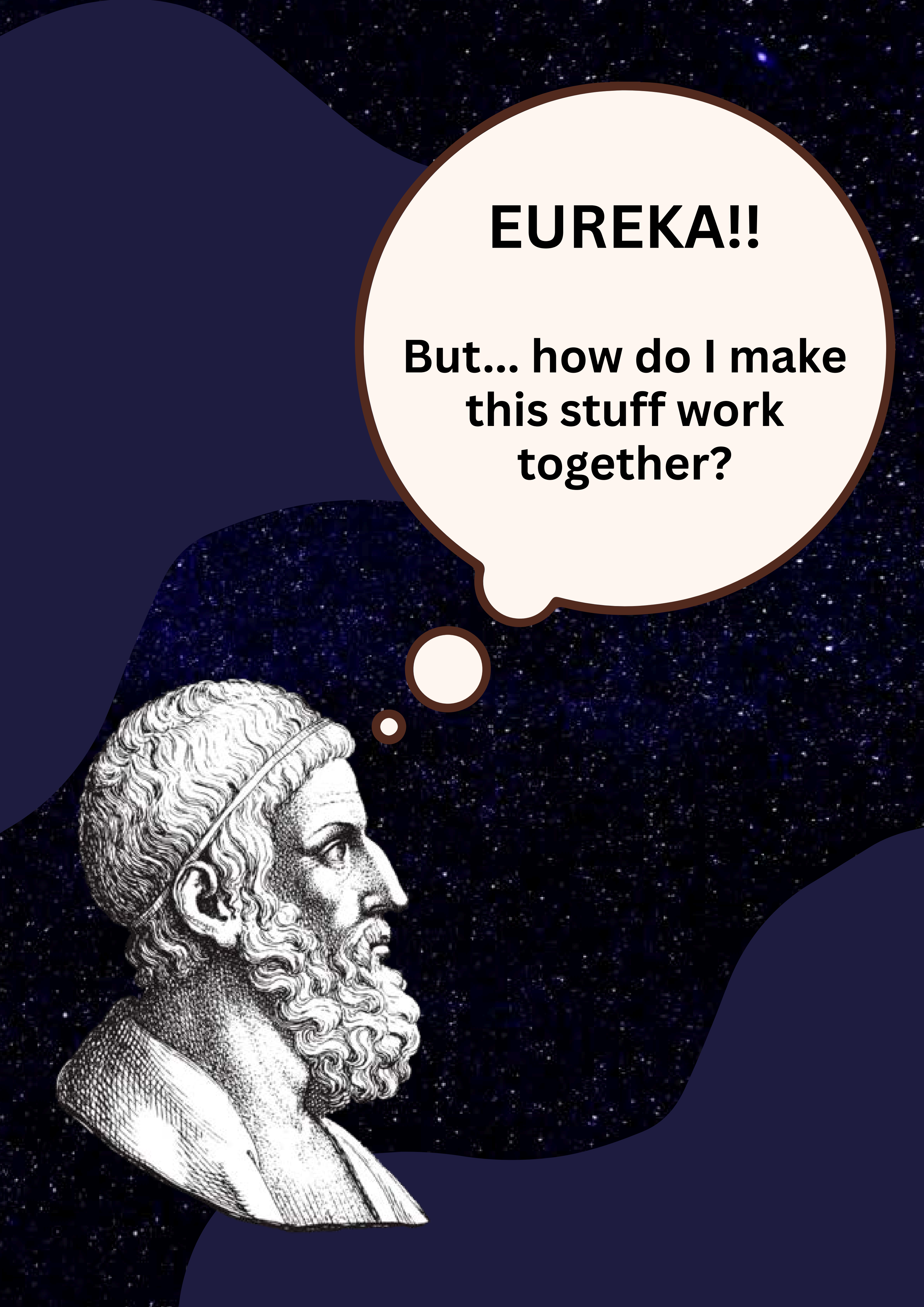
# Summary:


Coil #1
Coil #2
LiPo Battery
Ground Floor
Solar Charger IC
Board 1
H-bridge(s)
ESP32 S3 Mini
ESP32 CAM
MPU-6050
Gy-271 hmc5883l
Board 2
Camera
Solar Panel
Neodymium Magnet
Neodymium Magnet
Description
Power:
I2C and Power:
PWM and Power:
PWM:
Magnetic Field:
Wifi:


EUREKA!!
But... how do I make this stuff work together?

# 1st Step:

Download Arduino IDE

https://www.arduino.cc/en/software/

# 2nd Step:

Download some dependencies

- Go to “Preferences > Additional boards manager URL’s” and paste the following:

  https://espressif.github.io/arduino-esp32/package_esp32_index.json
  https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json

# 3rd Step:

Download the following codes from github

https://github.com/lumikalt/pic1-esp32/tree/main/final_version

https://github.com/lumikalt/pic1-esp32/tree/main/CameraWebServer

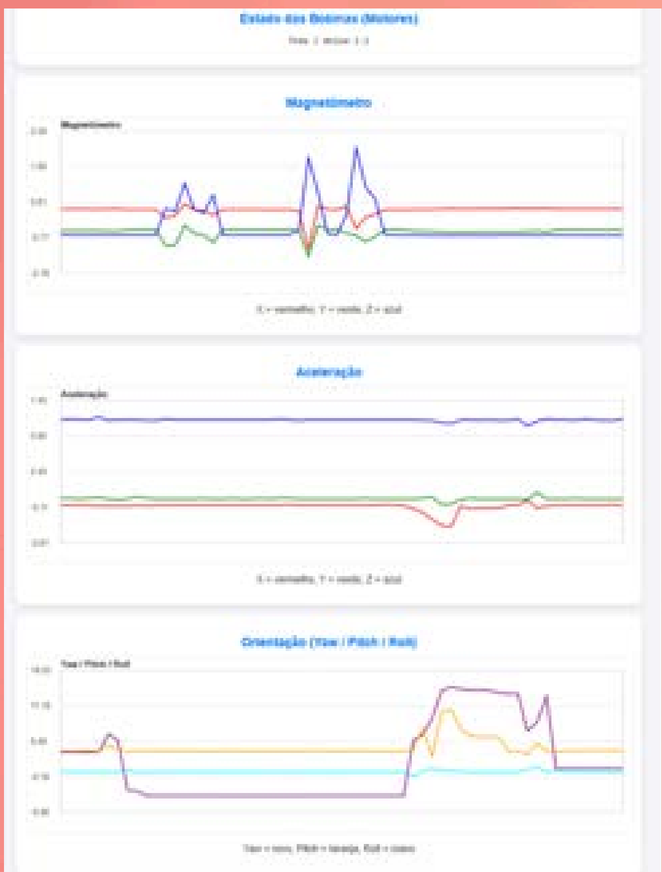
# 4th Step:

Open **CameraWebServer.ino** and **final_version.ino**

**NOTE:** Need to modify the lines **13-14** and **26-27** respectively

# 5th Step:

Get the IP adress from the Arduino IDE terminal in order to monitor and control the PocketQube as you please!




# Thank You!

**Our Website**

**Our Video**